INFOS-1101

Information systems definition

According to Britanica, Information system is an integrated set of components for collecting, storing, and processing data and for providing information, knowledge, and digital products.

An information system (IS) is a formal, sociotechnical, organizational system designed to collect, process, store, and distribute information. From a sociotechnical perspective, information systems are composed by four components: task, people, structure (or roles), and technology. Information systems can be defined as an integration of components for collection, storage and processing of data of which the data is used to provide information, contribute to knowledge as well as digital products that facilitate decision making. (Wikipedia).

At the most basic level, an information system (IS) is a set of components that work together to manage data processing and storage. Its role is to support the key aspects of running an organization, such as communication, record-keeping, decision making, data analysis and more. Companies use this information to improve their business operations, make strategic decisions and gain a competitive edge.

Takeaway Information

Many people think of information systems as computer-based technology. While information systems often incorporate computers to help manage data and achieve business objectives, they do not necessarily have to include computers.

There are different types of information systems that can serve a variety of purposes depending on an organization's needs. Examples include:

1. **Data warehouses**
   Data warehouses are data management systems that support analytics and other business intelligence activities. They consolidate and analyze data from a large variety of sources. Data warehouses can provide insight into businesses to help improve decision-making.
2. **Enterprise systems**
   Enterprise systems, also known as enterprise resource planning (ERP) systems, are integrated systems that combine all the hardware and software a business uses for different functions in

its operations. These organization-wide systems help information flow between departments and allow processes from different parts of the business to be integrated across a company.

3. **Expert systems**
   Expert systems use artificial intelligence to mimic human decision-making. The software uses human knowledge to solve problems that would typically require a person's expertise. Expert systems can be applied in areas such as medical diagnoses, accounting and coding.
4. **Geographic information systems**
   Geographic information systems (GIS) are tools that gather, organize, map and analyze data with a spatial component. GIS can improve analysis and decision-making by allowing users to visualize data on a map. Global information systems are a type of GIS that synthesize worldwide data.
5. **Office automation systems**
   Office automation systems combine communication technology, people and computers to help perform office activities, such as preparing written communication, printing, scheduling or creating reports.

Why study information systems

1. **Hands-on experience**: Information Technology & Systems is a subject area that demands practical experience. It will make you a desirable candidate in the graduate job market.

2. **Career flexibility:** Many different industries and sectors require the need for Information Technology & Systems graduates to help manage their systems. You could find yourself working for large firms, schools, administrative companies, the government, healthcare organisations, publishers or consultants.

3. **Transferable skills:** From IT to customer service, research to managerial, Information Technology & Systems students gain a host of valuable skills many employers are impressed by – even if they choose to work in a role that's not directly connected to the degree. A lot of jobs in the market today are based around work on a computer, so information-based skills are highly beneficial.

4. **Good starting salary:** According to graduate data, students from this subject area are likely to enjoy a good salary in a graduate job. Professional work following an Information Technology & Systems degree could be very technological focused. Companies pay good money to employ someone who can ensure their information is safe and secure.

Role of information systems

1. **Operational Efficiency:**
    - **a.** Automation: IS automate routine tasks, reducing errors and increasing productivity.
    - **b.** Process Improvement: By analyzing data, IS can identify bottlenecks and inefficiencies, leading to process optimization.
    - **c.** Resource Management: Efficient allocation of resources through systems like inventory management and HRIS.
2. **Decision Support:**
    a. Data Analysis: IS collect, process, and analyze vast amounts of data to provide valuable insights.
    b. Business Intelligence: Transforming data into actionable information for strategic decision-making.
    c. Predictive Analytics: Forecasting future trends and outcomes based on historical data.
3. **Customer Relationship Management (CRM):**
    a. Customer Interaction: Managing customer interactions across various channels (phone, email, social media).
    b. Sales and Marketing: Supporting sales and marketing efforts through customer data analysis and campaign management.
    c. Customer Service: Improving customer satisfaction through efficient issue resolution and support.
4. **Collaboration and Communication:**
    a. Intranet and Extranet: Facilitating internal and external communication and collaboration.
    b. Knowledge Management: Sharing information and expertise within the organization.
    c. Virtual Teams: Enabling remote work and collaboration among geographically dispersed teams.

5. **Competitive Advantage:**
    a. Innovation: IS can support the development of new products and services.
    b. Market Analysis: Identifying market trends and opportunities.
    c. Supply Chain Management: Optimizing supply chain operations for cost reduction and efficiency.

Components of Information Systems

Every information system includes several key components: hardware, software, telecommunications, people and data. Hardware refers to the physical pieces of the information system; software is the programming that controls the information system; telecommunication transmits information through the system; humans manage and interact with the information system; and data is information stored within and processed by the system.

Hardware

The hardware component of an information system comprises the physical elements of the system. People can touch and feel pieces of hardware. These mechanisms, equipment and wiring allow systems like computers, smartphones and tablets to function.

Input and output devices are essential pieces of technology that allow humans to interact with computers and other information systems. Keyboards, mice, microphones and scanners are all examples of input devices. And output devices might include printers, monitors, speakers and sound and video cards.

Pieces of hardware including microprocessors, hard drives, electric power supply units, and removable storage also allow computers to store and process data.

Software

Software are the intangible programs that manage information system functions, including input, output, processing and storage.

System software – such as the MacOS or Microsoft Windows operating systems – provides a base for application software to run.

Application software operates programs geared toward particular uses in information systems. For example, word processing applications are used to create and edit text documents. Graphical user interface (GUI) software is among the most common application software; it presents the information stored in computers and allows users to interact with computers through digital graphics – such as icons, buttons and scroll bars – rather than through text-based commands.

Software can be either open source or closed source. Open source software coding is publicly available for users and programmers to manipulate, whereas closed source software is proprietary.

Telecommunications

Telecommunications systems connect computer networks and allow information to be transmitted through them. Telecommunications networks also allow computers and storage services to access information from the cloud.

There are a number of methods telecommunications networks use to convey information. Coaxial cables and fiber optic cables are used by telephone, internet and cable providers to transmit data, video and audio messages.

Local-area networks (LANs) connect computers to create computer networks in a designated space, like a school or home. Wide-area networks (WANs) are collections of LANs that facilitate data-sharing across large areas. A virtual private network (VPN) allows a user to protect their online privacy by encrypting data on public networks.

Microwaves and radio waves can also be used to transmit information in telecommunications networks.

Data

Data are intangible, raw facts that are stored, transmitted, analyzed and processed by other components of information systems. Data are often stored as numerical facts, and they represent quantitative or qualitative information.

Data can be stored in a database or data warehouse, in a form that best suits the organization using it.

Databases house collections of data that can be queried or retrieved for specific purposes. Databases allow users to perform fundamental operations, such as storage and retrieval. Data warehouses, on the other hand, store data from multiple sources for analytical purposes. They allow users to assess an organization or its operations.

Human Resources

Human resources are a crucial part of information systems. The human component of information systems encompasses the qualified people who influence and manipulate the data, software and processes in information systems. Humans involved in information systems may include business analysts, information security analysts or system analysts.

Business analysts work to elevate an organization’s operations and processes. They often focus on improving efficiency and productivity or streamlining distribution. Information security analysts work to prevent data breaches and cybersecurity attacks. And system analysts use information technology to help organizations optimize their user experiences with programs.